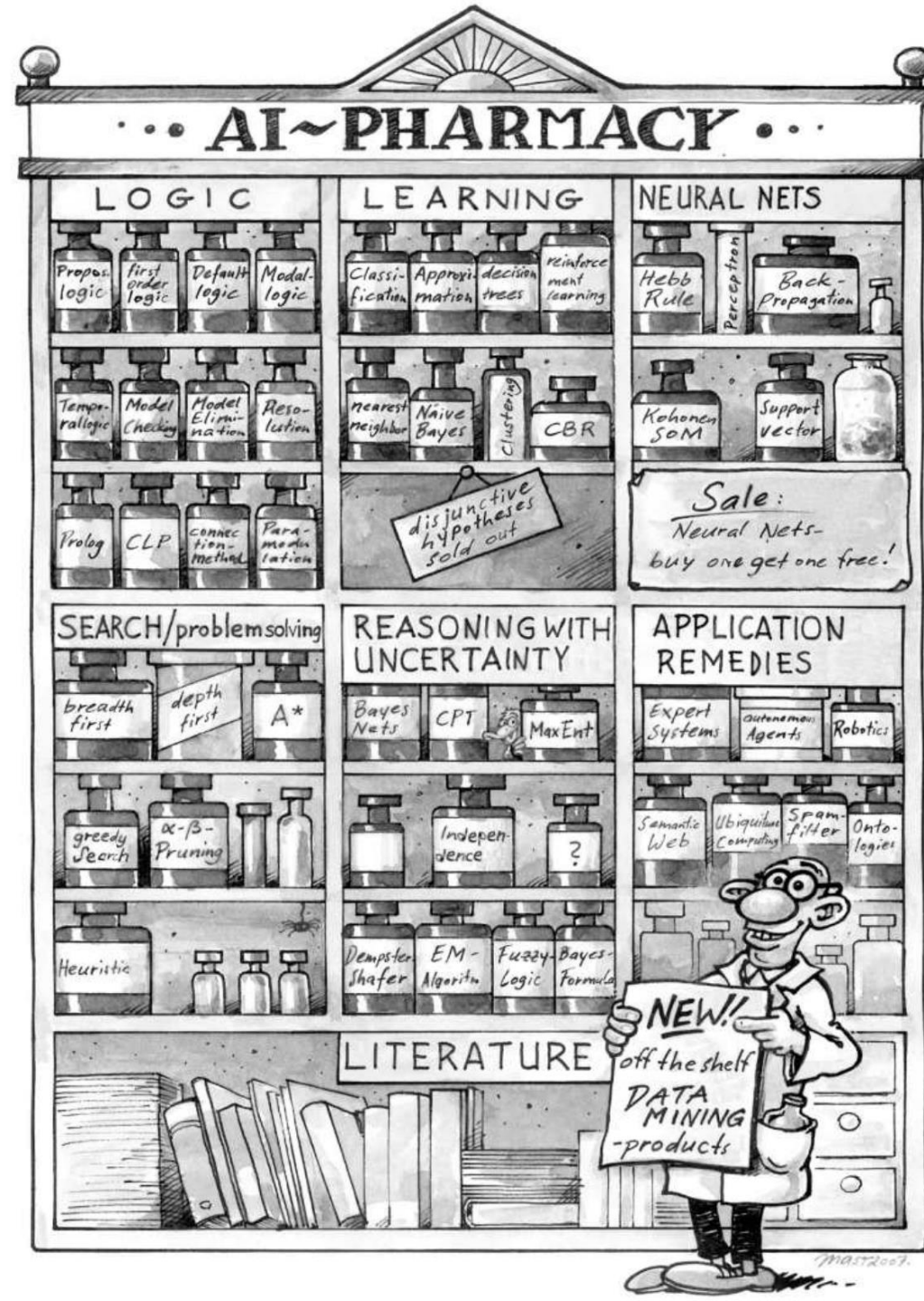


Logical Agents

KECERDASAN
KOMPUTASIONAL (N)



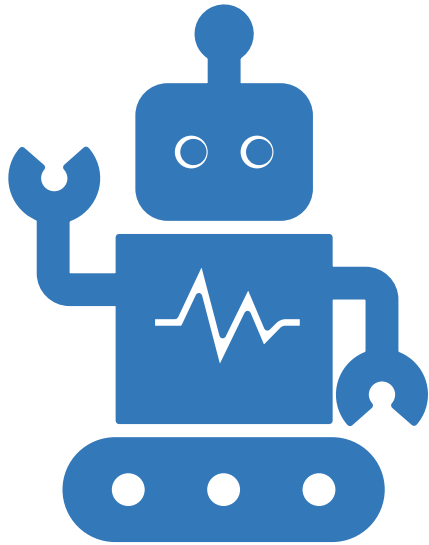


Logical Agents

Disebut juga *Knowledge based agents*

- Membuat **representasi** dari complex world
- Melakukan **inferensi** dari representasi menjadi representasi baru
- Membuat **kesimpulan** aksi berdasarkan representasi-representasi baru

Knowledge-based Agents



It is how AI approaches **intelligence**.

Knowledge-based agents can:

- **accept new tasks** in the form of explicitly described goals;
- **achieve competence** quickly by being told or learning new knowledge about the environment;
- **adapt to changes** in the environment by updating the relevant knowledge

Knowledge-based Agents

The central component: **knowledge base**, or KB.

A knowledge base: a set of **sentences**.

Each sentence is expressed in a language: a **knowledge representation language**.

To add new sentences to the knowledge base and to query what is known:

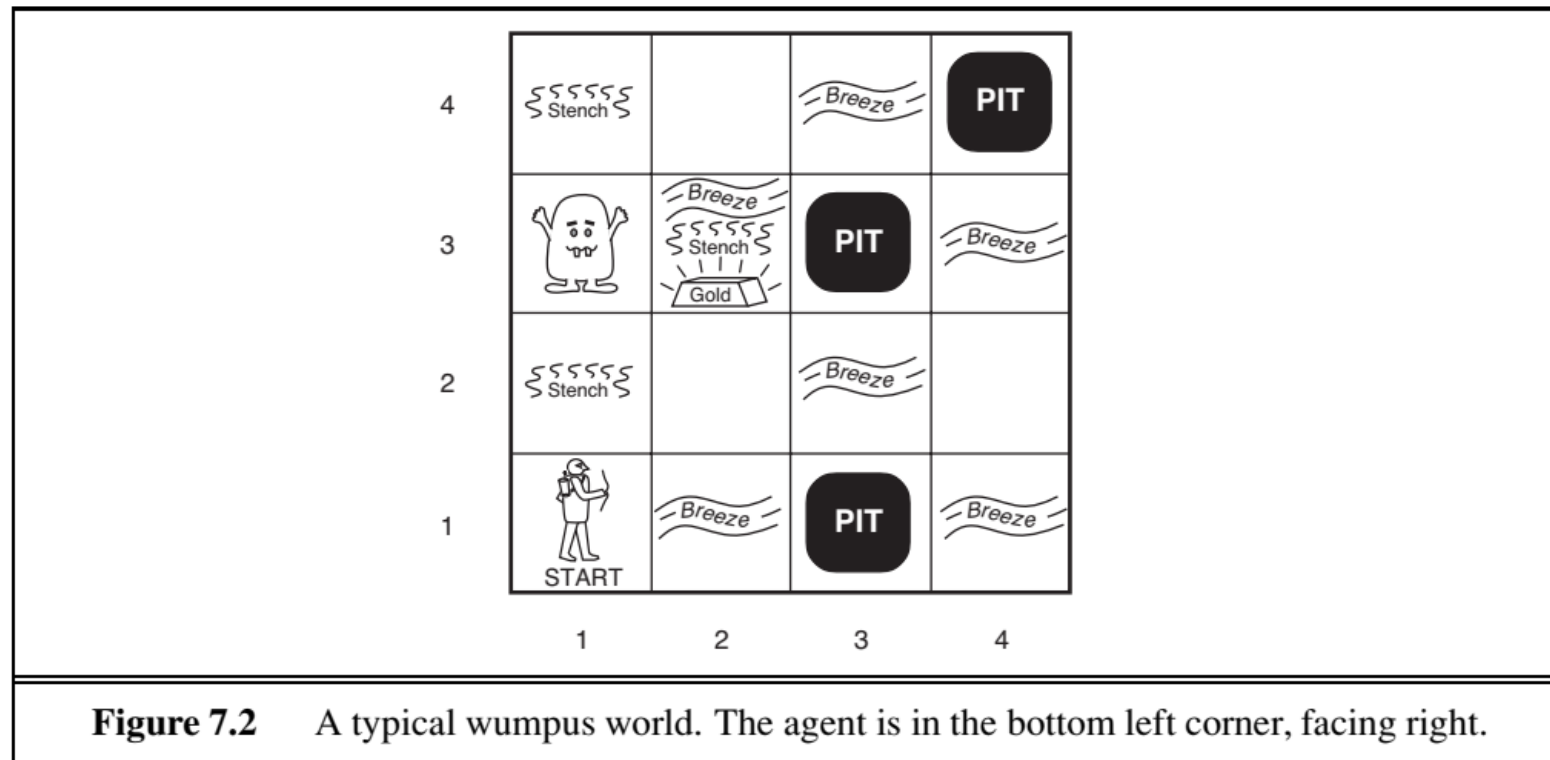
- Operations are TELL and ASK, respectively.
- Both operations may involve **inference**—that is, deriving new sentences from old.

Knowledge-based Agents

```
function KB-AGENT(percept) returns an action  
  persistent: KB, a knowledge base  
               t, a counter, initially 0, indicating time  
  
  TELL(KB, MAKE-PERCEPT-SENTENCE(percept, t))  
  action  $\leftarrow$  ASK(KB, MAKE-ACTION-QUERY(t))  
  TELL(KB, MAKE-ACTION-SENTENCE(action, t))  
  t  $\leftarrow$  t + 1  
  return action
```

Figure 7.1 A generic knowledge-based agent. Given a percept, the agent adds the percept to its knowledge base, asks the knowledge base for the best action, and tells the knowledge base that it has in fact taken that action.

The Wumpus World



The Wumpus World

A cave consisting of rooms connected by passageways

Wumpus: a beast that eats anyone who enters its room

The wumpus can be shot by an agent, but the agent has only one arrow.

Some rooms contain bottomless pits that will trap anyone who wanders into these rooms.

The only mitigating feature of this bleak environment is the possibility of finding a heap of gold.

The Wumpus World: PEAS definition

Performance measure

- +1000: gold
- -1000: fall into a pit or being eaten by the Wumpus
- -1: for each action taken
- -10: for using up the arrow
- The game ends either when the agent dies or when the agent climbs out of the cave.

Environment

Actuators

Sensors

The Wumpus World: PEAS definition

Performance measure

Environment

- A 4×4 grid of rooms.
- The agent always starts in the square labeled [1,1], facing to the right.
- The locations of the gold and the wumpus are chosen randomly, with a uniform distribution, from the squares other than the start square.
- Each square other than the start can be a pit, with probability 0.2.

Actuators

Sensors

The Wumpus World: PEAS definition

Performance measure

Environment

Actuators

- The agent can move *Forward*, *TurnLeft* by 90°, or *TurnRight* by 90°.
- The agent dies if it enters a square containing a pit or a live wumpus.
- If an agent tries to move forward and bumps into a wall, then the agent does not move.
- The action *Grab* is used to pick up the gold if it is in the same square as the agent.
- The action *Shoot* is used to fire an arrow in a straight line in the direction the agent is facing.
- The action *Climb* can be used to climb out of the cave, but only from square [1,1].

Sensors

The Wumpus World: PEAS definition

Performance measure

Environment

Actuators

Sensors

- In the square containing the wumpus and in the directly (not diagonally) adjacent squares, the agent will perceive a **Stench**.
- In the squares directly adjacent to a pit, the agent will perceive a **Breeze**.
- In the square where the gold is, the agent will perceive a **Glitter**.
- When an agent walks into a wall, it will perceive a **Bump**.
- When the wumpus is killed, it emits a **Scream** that can be perceived anywhere in the cave.
- Percept example: [Stench, Breeze, None, None, None]

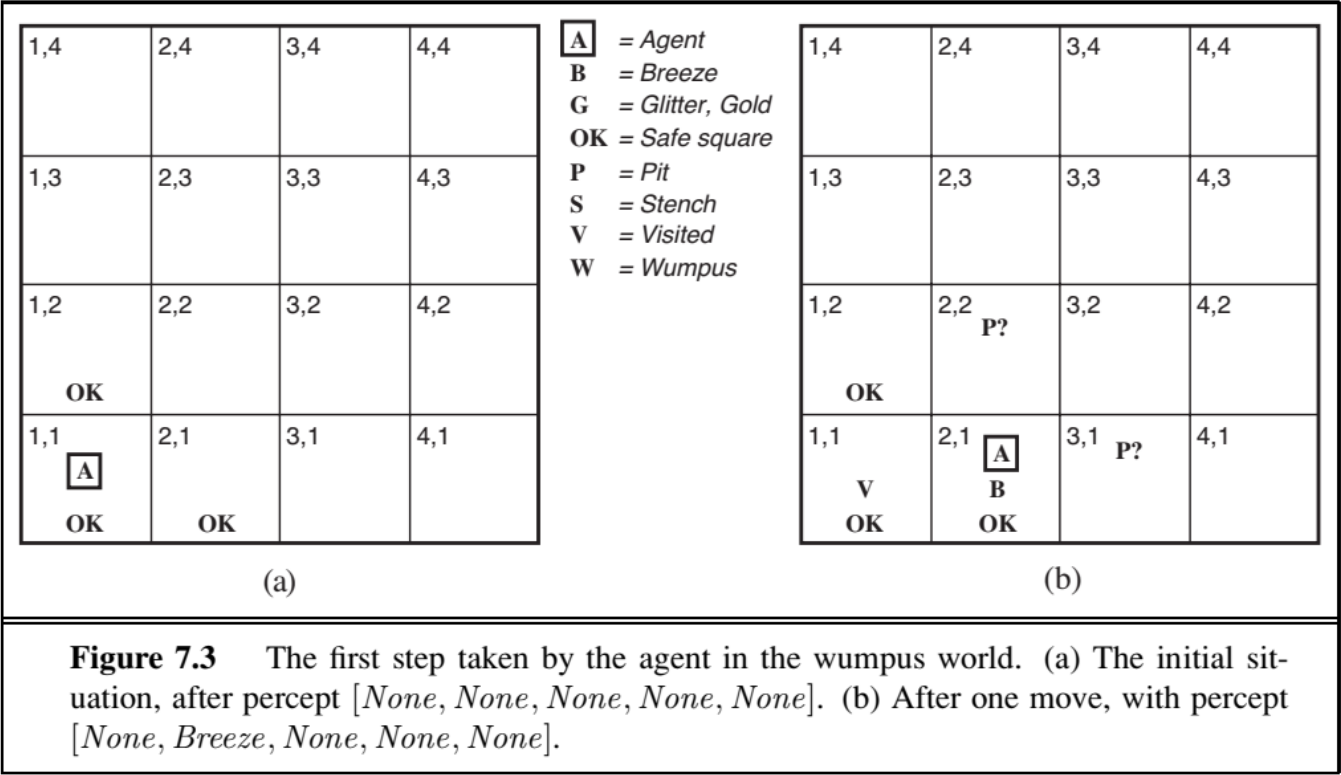
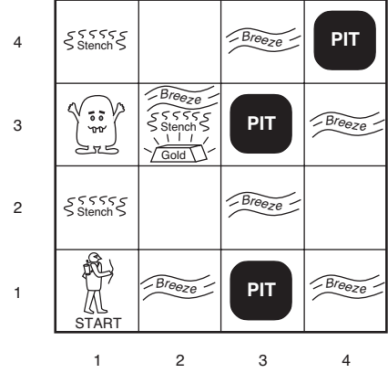
The Wumpus Environment Characteristics

Discrete, static, and single-agent.

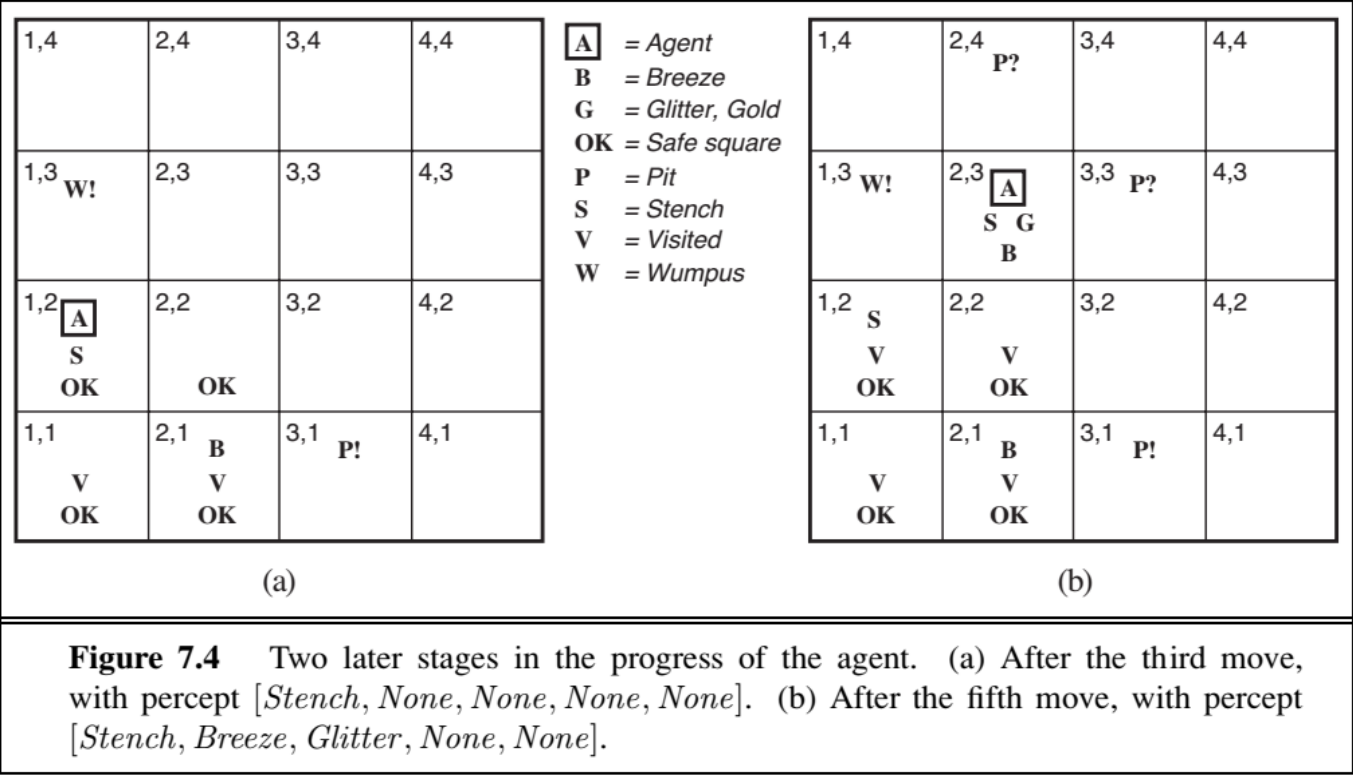
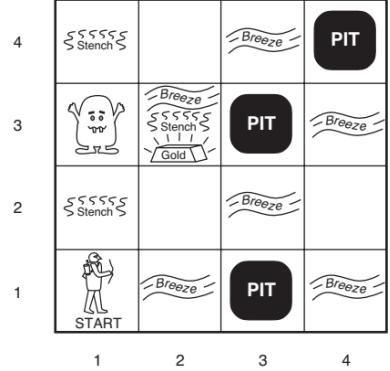
Sequential: rewards may come only after many actions are taken.

Partially observable: some aspects of the state are not directly perceivable, e.g. the agent's location, the wumpus's state of health, and the availability of an arrow.

A knowledge-based wumpus agent exploring the environment shown in Figure 7.2.



A knowledge-based wumpus agent exploring the environment shown in Figure 7.2.



Logical Reasoning

Note that in each case for which the agent draws a conclusion from the available information, that conclusion is guaranteed to be correct if the available information is correct.

This is a fundamental property of **logical reasoning**.

Terminologies

Knowledge bases consist of sentences, expressed according to the **syntax** of the representation language

- E.g., in arithmetic, “ $x + y = 4$ ” is a well-formed sentence, whereas “ $x3y+ =$ ” is not.

A logic must also define the **semantics** or meaning of sentences.

- E.g., the semantic for arithmetic specifies that the sentence “ $x + y = 4$ ” is true in a world where x is 2 and y is 2, but false in a world where x is 1 and y is 1.

If a sentence α is true in model m , then m **satisfies** α or m is a **model** of α .

- The notation $M(\alpha)$ is the set of all models of α .

The relation of **logical entailment** between sentences: a sentence follows logically from another sentence

$$\alpha \models \beta$$

Logical Entailment Sample

Consider the following situation: the agent has detected nothing in [1,1] and a breeze in [2,1].

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2 P?	3,2	4,2
1,1 V OK	2,1 A B OK	3,1 P?	4,1

These percepts, combined with the agent's knowledge of the rules of the wumpus world, constitute the KB.

Logical Entailment Sample

1,4	2,4	3,4	4,4
1,3	2,3	3,3	4,3
1,2 OK	2,2 P?	3,2	4,2
1,1 V OK	2,1 A B OK	3,1 P?	4,1

The agent's percepts: nothing in [1,1] and a breeze in [2,1].

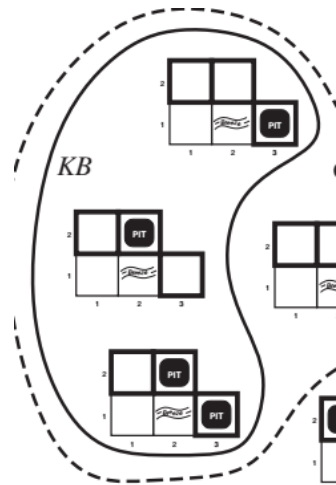
The agent is interested in whether the adjacent squares [1,2], [2,2], and [3,1] contain pits.

Each of the three squares might or might not contain a pit, so there are $2^3 = 8$ possible models.

The knowledge base (KB) is a set of sentences.

The KB is false in models that contradict what the agent knows

- The KB is false in any model in which [1,2] contains a pit, because there is no breeze in [1,1].
- There are just three models in which the KB is true.



Logical Entailment Sample

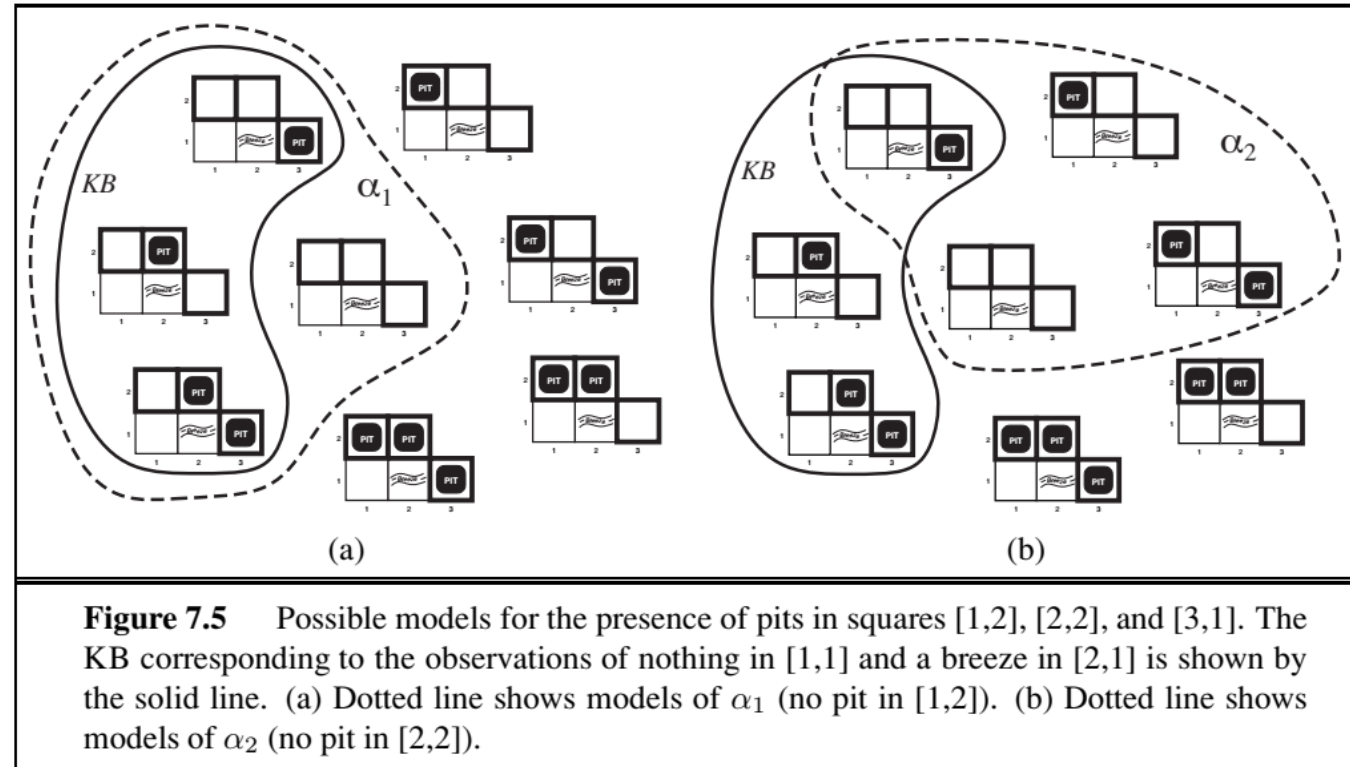
Let's consider two possible conclusions:

α_1 = "There is no pit in [1,2]."

α_2 = "There is no pit in [2,2]."

By inspection, we see the following: in every model in which KB is true, α_1 is also true.

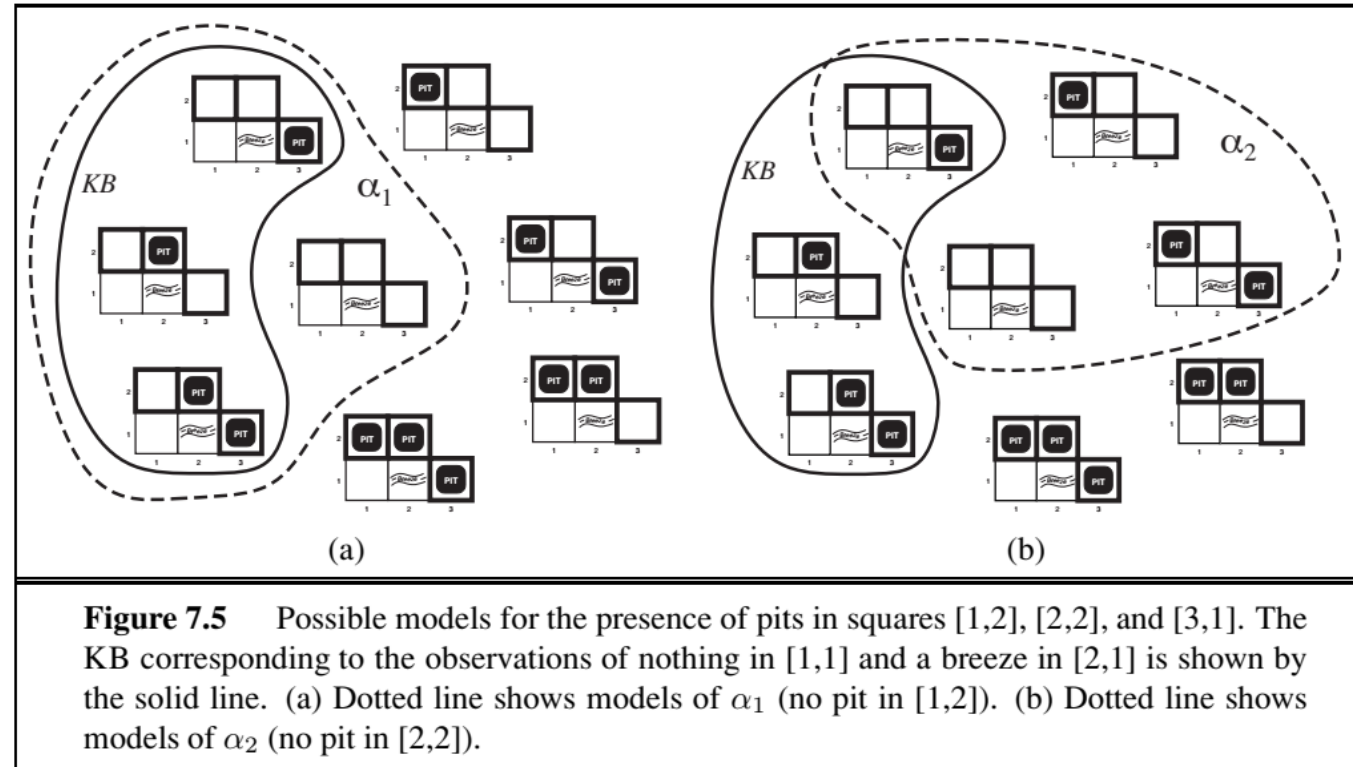
Hence, $KB \models \alpha_1$: there is no pit in [1,2].



Logical Entailment Sample

We can also see that in some models in which KB is true, α_2 is false.

Hence, $KB \not\models \alpha_2$: the agent cannot conclude that there is no pit in [2,2].



Logical Inference

Entailment can be applied to derive conclusions or carry out **logical inference**.

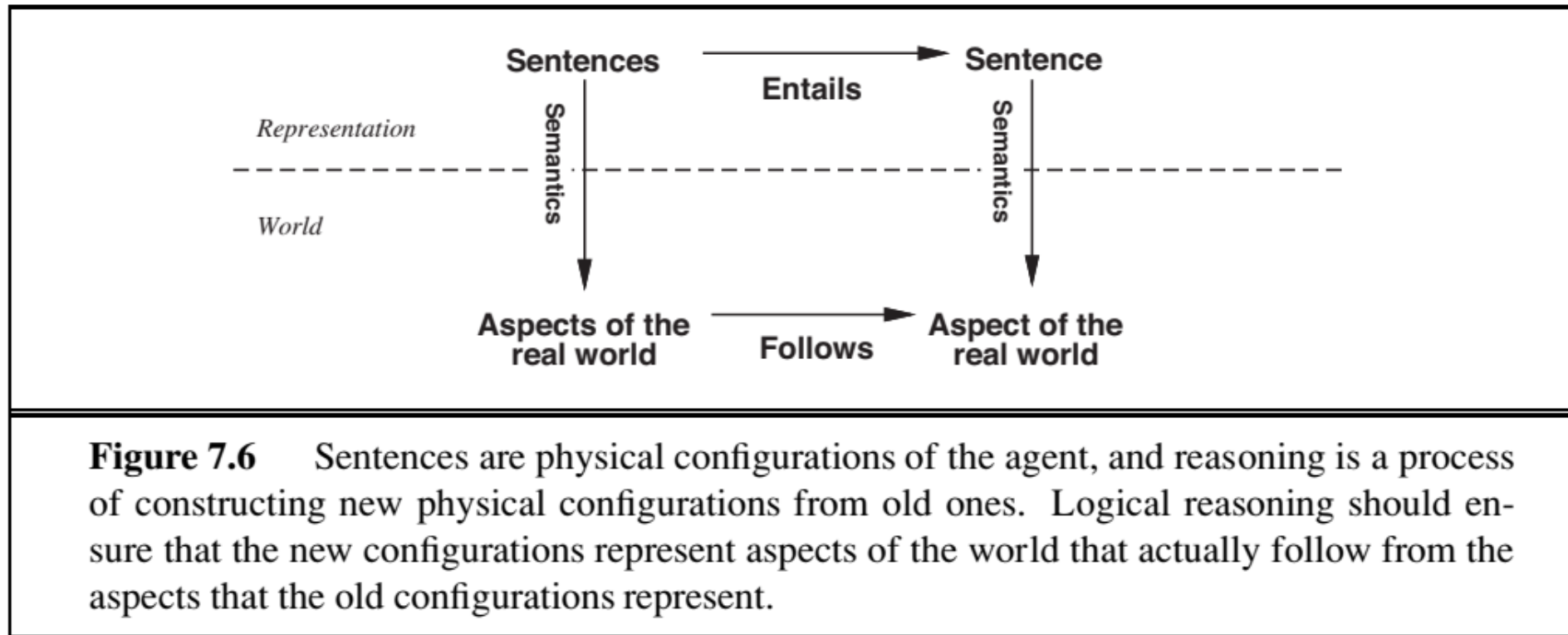
The inference algorithm illustrated before is called **model checking**, because it **enumerates all possible models to check that α is true in all models in which KB is true**, that is, that $M(KB) \subseteq M(\alpha)$.

An inference algorithm that derives only entailed sentences is called **sound** or **truth preserving**.

- Model checking, when it is applicable, is a sound procedure.

An inference algorithm is **complete** if it can derive any sentence that is entailed.

Logical Reasoning



Propositional Logic

A VERY SIMPLE
LOGIC

Terminologies in Propositional Logic

Syntax: the allowable sentences

Proposition symbol: a proposition that can be true or false, e.g: P , Q , R , W_{12} , *North*, *True*, *False*.

Complex sentences: construction of simpler sentences using parentheses and **logical connectives** (not, and, or, implies, if and only if).

Logical Connectives

NEGATION	$\neg$ (not). A sentence such as $\neg W_{1,3}$ is called the negation of $W_{1,3}$. A literal is either an atomic sentence (a positive literal) or a negated atomic sentence (a negative literal).
LITERAL	
CONJUNCTION	$\wedge$ (and). A sentence whose main connective is $\wedge$, such as $W_{1,3} \wedge P_{3,1}$, is called a conjunction ; its parts are the conjuncts . (The $\wedge$ looks like an “A” for “And.”)
DISJUNCTION	$\vee$ (or). A sentence using $\vee$, such as $(W_{1,3} \wedge P_{3,1}) \vee W_{2,2}$, is a disjunction of the disjuncts $(W_{1,3} \wedge P_{3,1})$ and $W_{2,2}$. (Historically, the $\vee$ comes from the Latin “vel,” which means “or.” For most people, it is easier to remember $\vee$ as an upside-down $\wedge$.)
IMPLICATION	$\Rightarrow$ (implies). A sentence such as $(W_{1,3} \wedge P_{3,1}) \Rightarrow \neg W_{2,2}$ is called an implication (or conditional). Its premise or antecedent is $(W_{1,3} \wedge P_{3,1})$, and its conclusion or consequent is $\neg W_{2,2}$. Implications are also known as rules or if–then statements. The implication symbol is sometimes written in other books as $\supset$ or $\rightarrow$.
PREMISE	
CONCLUSION	
RULES	
BICONDITIONAL	$\Leftrightarrow$ (if and only if). The sentence $W_{1,3} \Leftrightarrow \neg W_{2,2}$ is a biconditional . Some other books write this as $\equiv$.

Diskusi Kelompok

Diskusikan bersama 1 orang temanmu

Diketahui proposisi *atomic* x_1, x_2, x_3 dimana x_i merupakan notasi untuk kalimat “pegawai i sedang bekerja”. Buatlah proposisi sesuai kondisi berikut.

1. Salah satu dari pegawai 1, 2 dan 3 sedang bekerja.
2. Terdapat dua pegawai dari pegawai 1, 2, dan 3 yang sedang bekerja.
3. Tidak semua pegawai sedang bekerja.

Jika sudah, diskusikan jawaban kelompokmu dengan kelompok lain

Semantics

Semantics: the **rules** for determining the truth of a sentence with respect to a particular **model**.

A **model**: fixes the truth value—true or false—for every proposition symbol.

- For example, if the sentences in the knowledge base make use of the proposition symbols $P_{1,2}$, $P_{2,2}$, and $P_{3,1}$, then one possible model is

$$m_1 = \{P_{1,2} = \text{false}, P_{2,2} = \text{false}, P_{3,1} = \text{true}\} .$$

The rules: expressed with **truth tables** that specify the truth value of a complex sentence

Truth Tables

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>true</i>
<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>
<i>true</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>
<i>true</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>

Figure 7.8 Truth tables for the five logical connectives. To use the table to compute, for example, the value of $P \vee Q$ when P is true and Q is false, first look on the left for the row where P is *true* and Q is *false* (the third row). Then look in that row under the $P \vee Q$ column to see the result: *true*.

In $m_1 = \{P_{1,2} = \text{false}, P_{2,2} = \text{false}, P_{3,1} = \text{true}\}$, determine the value of the sentence: $\neg P_{1,2} \wedge (P_{2,2} \vee P_{3,1})$

Truth Tables

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>true</i>
<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>
<i>true</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>
<i>true</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>

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In propositional logic, is it True or False: The sentence “5 is odd implies Tokyo is the capital of Japan”

Truth Tables

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>true</i>
<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>false</i>
<i>true</i>	<i>false</i>	<i>false</i>	<i>false</i>	<i>true</i>	<i>false</i>	<i>false</i>
<i>true</i>	<i>true</i>	<i>false</i>	<i>true</i>	<i>true</i>	<i>true</i>	<i>true</i>

Figure 7.8 Truth tables for the five logical connectives. To use the table to compute, for example, the value of $P \vee Q$ when P is true and Q is false, first look on the left for the row where P is *true* and Q is *false* (the third row). Then look in that row under the $P \vee Q$ column to see the result: *true*.

In propositional logic, is it True or False: The sentence “5 is even implies Sam is smart”

Example in Wumpus World to derive $\neg P_{1,2}$ (there is no pit in [1,2])

Symbols: $P_{x,y}$ is true if there is a pit in $[x, y]$.
 $W_{x,y}$ is true if there is a wumpus in $[x, y]$, dead or alive.
 $B_{x,y}$ is true if the agent perceives a breeze in $[x, y]$.
 $S_{x,y}$ is true if the agent perceives a stench in $[x, y]$.

- There is no pit in [1,1]:

$$R_1 : \neg P_{1,1} .$$

- A square is breezy if and only if there is a pit in a neighboring square. This has to be stated for each square; for now, we include just the relevant squares:

$$R_2 : B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1}) .$$

$$R_3 : B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1}) .$$

- The preceding sentences are true in all wumpus worlds. Now we include the breeze percepts for the first two squares visited in the specific world the agent is in, leading up to the situation in Figure 7.3(b).

$$R_4 : \neg B_{1,1} .$$

$$R_5 : B_{2,1} .$$

Example in Wumpus World to derive $\neg P_{1,2}$ (there is no pit in [1,2])

Goal: decide whether $KB \models \alpha$ for some sentence α .

- For example, is $\neg P_{1,2}$ entailed by the Wumpus World KB?

Algorithm: model checking $\rightarrow$ enumerate the models, and check that α is true in every model in which KB is true.

Currently there are 7 symbols: ?

Number of possible models: ?

- There is no pit in [1,1]:
 $R_1 : \neg P_{1,1} .$
- A square is breezy if and only if there is a pit in a neighboring square. This has to be stated for each square; for now, we include just the relevant squares:
 $R_2 : B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1}) .$
 $R_3 : B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1}) .$
- The preceding sentences are true in all wumpus worlds. Now we include the breeze percepts for the first two squares visited in the specific world the agent is in, leading up to the situation in Figure 7.3(b).
 $R_4 : \neg B_{1,1} .$
 $R_5 : B_{2,1} .$

Example in Wumpus World to derive $\neg P_{1,2}$ (there is no pit in [1,2])

$B_{1,1}$	$B_{2,1}$	$P_{1,1}$	$P_{1,2}$	$P_{2,1}$	$P_{2,2}$	$P_{3,1}$	R_1	R_2	R_3	R_4	R_5	KB
false	false	false	false	false	false	false	true	true	true	true	false	false
false	false	false	false	false	false	true	true	true	false	true	false	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
false	true	false	false	false	false	false	true	true	false	true	true	false
false	true	false	false	false	false	true	true	true	true	true	true	<u>true</u>
false	true	false	false	false	true	false	true	true	true	true	true	<u>true</u>
false	true	false	false	false	true	true	true	true	true	true	true	<u>true</u>
false	true	false	false	true	false	false	true	false	false	true	true	false
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
true	true	true	true	true	true	true	false	true	true	false	true	false

Figure 7.9 A truth table constructed for the knowledge base given in the text. KB is true if R_1 through R_5 are true, which occurs in just 3 of the 128 rows (the ones underlined in the right-hand column). In all 3 rows, $P_{1,2}$ is false, so there is no pit in [1,2]. On the other hand, there might (or might not) be a pit in [2,2].

Latihan

1,4	2,4	3,4	4,4
1,3 W!	2,3	3,3	4,3
1,2 A S OK	2,2 OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1

A = Agent
B = Breeze
G = Glitter, Gold
OK = Safe square
P = Pit
S = Stench
V = Visited
W = Wumpus

Diketahui *agent* telah sampai pada kondisi seperti pada gambar di samping: tidak ada apa-apa pada [1,1], ada input berupa bau (*stench*) pada [1,2], dan belum tahu isi dari [1,3], [2,2], dan [3,1]. Masing-masing dari posisi tsb dapat berisi sebuah lubang (pit) dan maksimal satu posisi dapat berisi monster (Wumpus).

Tentukan kondisi yang mungkin (sebanyak 32 *possible worlds*).

Tandai kondisi dimana KB bernilai benar dan dimana masing-masing kalimat berikut benar:

α_2 = "There is no pit in [2,2]."

α_3 = "There is a wumpus in [1,3]."

Dengan demikian tunjukkan bahwa $KB \models \alpha_2$ dan $KB \models \alpha_3$.

Next Week: Propositional Theorem Proving

PROOF
CONSTRUCTION
WITHOUT
CONSULTING
MODELS